

# Project Description: Optimizing Cart Abandonment Email Timing

This project was initiated with the objective of rigorously testing which specific timing for sending cart abandonment emails—3 hours, 6 hours, or 18 hours—leads to the highest return-to-purchase rate. The overarching goal was to provide data-driven recommendations for e-commerce strategies to effectively recover abandoned carts and boost conversion.

The project commenced with the crucial step of **data loading**, where the relevant dataset containing user return information was imported. Following this, a **power analysis** was conducted to ensure the study had sufficient statistical power to detect meaningful differences. This analysis determined a minimum required sample size of 156. The actual dataset used for the analysis contained three distinct email timing groups: '1hr', '6hr', and '24hr', each consisting of 1000 users. It is important to note the discrepancy between the objective's stated timings (3 hr, 6 hrs, 18 hrs) and the actual data's group labels (1hr, 6hr, 24hr).

The initial **return rates by group** were calculated, providing an empirical overview of performance:

- The '1hr' group showed a return-to-purchase rate of 26.2%.
- The '24hr' group achieved a rate of 29.0%.
- The '6hr' group demonstrated the highest observed return rate at 30.9%.

To statistically evaluate whether these observed differences in return rates were significant, an **Independence test using Chi-Square** was performed. A contingency table was constructed to summarize the counts of users who returned (True) or did not return (False) for each timing group:

| returned | False | True |
|----------|-------|------|
| group    |       |      |
| 1hr      | 738   | 262  |
| 24hr     | 710   | 290  |
| 6hr      | 691   | 309  |

The Chi-Square test yielded a chi-squared statistic of 5.4635 with a p-value of 0.0651. At a conventional significance level ( $\alpha$ ) of 0.05, since the p-value (0.0651) is greater than  $\alpha$ , the

project **failed to reject the null hypothesis**. This statistical outcome suggests that, based on this dataset, the return-to-purchase rate is **independent of the email timing group** (1hr, 6hr, 24hr). In simpler terms, the observed differences in return rates across the groups are not statistically significant and could be due to random chance.

Beyond the primary analysis of return rates, the project also delved into the **delay to return** for those users who did make a purchase. A **Normality test** (Shapiro-Wilk) was conducted for the `delay_to_return_hr` data within each group. The results indicated that the p-values for all three groups ('1hr', '6hr', '24hr') were 0.0000, leading to the conclusion that the data in each group is **not normally distributed** (rejecting the null hypothesis of normality). This finding is crucial as it dictates the choice of subsequent statistical tests.

Given that the time taken for users to return was found to follow a non-normal (likely right-skewed, exponential) distribution, a **non-parametric test** was required to compare the central tendencies of `delay_to_return_hr` across groups. The **Kruskal-Wallis H-test** was applied, yielding an H-statistic of 433.2935 with an extremely low p-value of 0.0000. This highly significant p-value indicates that there are **significant differences in the delay-to-return across the groups**.

To identify which specific groups differed from each other in terms of delay-to-return, a **Post-hoc analysis using the Mann-Whitney U test** was performed for all pairwise comparisons. The results showed statistically significant differences for all pairs:

- 1hr vs. 6hr: raw p = 0.0000, adjusted p = 0.0000
- 1hr vs. 24hr: raw p = 0.0000, adjusted p = 0.0000
- 6hr vs. 24hr: raw p = 0.0000, adjusted p = 0.0000

These results confirm that the distributions of delay-to-return are significantly different between every pair of email timing groups.

Finally, to quantify the magnitude of these differences, **Cohen's d effect size** was calculated for the `delay_to_return_hr` between the groups:

- Cohen's d for 6hr vs. 1hr: 1.1495
- Cohen's d for 24hr vs. 1hr: 1.3518
- Cohen's d for 24hr vs. 6hr: 0.9589

All these Cohen's d values are greater than 0.8, indicating a **large difference between groups** in terms of the delay taken for users to return to complete their purchase. This means that while email timing may not significantly influence *whether* a user returns (as per the Chi-Square test), it has a substantial impact on *how quickly* they return once they decide to.

## Conclusions:

1. **Return-to-Purchase Rate:** Despite observing that the 6-hour email timing had the highest return rate (30.9%), the **Chi-Square test indicated no statistically significant**

**difference in return-to-purchase rates** across the 1-hour, 6-hour, and 24-hour email timing groups. This means that, based on this dataset and analysis, altering the email timing within these intervals does not definitively change the likelihood of a customer returning to complete a purchase.

2. **Delay to Return:** Although the timing did not significantly affect the *occurrence* of a return, it had a **profound and statistically significant impact on the speed of return**. The non-parametric Kruskal-Wallis test and subsequent Mann-Whitney U post-hoc tests revealed significant differences in the `delay_to_return_hr` across all groups. The large Cohen's d effect sizes further emphasize these substantial differences, suggesting that while all timings might eventually lead to similar overall return rates, some timings are much more effective at prompting quicker conversions from those who do return.

In summary, while the email timing strategy did not yield a statistically significant difference in the *overall return-to-purchase rate* in this study, it strongly influenced the *speed at which customers returned to complete their purchase*. Businesses may consider this insight when planning their cart abandonment strategies, prioritizing quicker conversions even if the ultimate return rate remains similar across tested timings.